

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. (CL) Examination

November 2012

CL303 Geotechnical Engineering-I

Date: 28.11.2012, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Enlist the limitations of soil engineering. [03]

(b) Differentiate between compaction and consolidation. [04]

Q - 2 (a) Explain spring analogy for primary consolidation, with neat sketch. [05]

(b) During laboratory test on a clay sample of thickness 25 mm, drained at top only, 50% consolidation occurred in 11 minutes. Find the time required for the corresponding clay layer in the field 2 m thick and drained at top and bottom to undergo 70% consolidation. Assume $T_{50} = 0.197$ and $T_{70} = 0.405$. [05]

(c) In the Standard compaction test, following results are obtained: [04]

Water content, %	8.80	9.60	12.10	15.00	17.90	21.00	24.20
Bulk density, kN/m^3	12.62	13.19	15.60	17.20	17.50	17.00	16.60

Plot compaction curve and obtain OMC and MDD.

OR

Q - 2 (a) Define consolidation & give reasons of occurring consolidation. Explain initial, primary and secondary consolidation. [05]

(b) When consolidation test was performed, the void ratio of the specimen which was 1.070 under the effective pressure of 220 kN/m^2 , changed to 0.980 when the pressure was increased to 430 kN/m^2 . Calculate (i) a_v , (ii) C_c , (iii) m_v . Find the settlement of foundation (S_c) when resting on 10 m thick clay layer. The increase in pressure is 10 kN/m^2 . [05]

(c) Explain effect of compaction on soil structure and permeability. [04]

Q - 3 (a) Write steps to find out pre-consolidation pressure. [02]

(b) Explain in detail the test to obtain shear parameters of the stiff clays or granular soil, with neat sketch of experimental set up. [07]

(c) A series of shear tests were performed on a soil. Each test was carried out until the sample sheared and the principal stresses for each test were given in the following table. Plot the Mohr's circle to get shear parameters for total stress condition and [05]

effective shear stress condition, if pore water pressure is 100 kN/m^2 ;

Test No.	$\sigma_1 \text{ (kN/m}^2\text{)}$	$\sigma_d \text{ (kN/m}^2\text{)}$
1	600	400
2	900	600
3	1200	800

OR

- Q - 3 (a) Differentiate between over consolidated clay and normally consolidated clay. [02]
 (b) Enlist the types of triaxial tests. Explain all of them in detail with sketches. [07]
 (c) Two drained triaxial tests are performed on the material. In the first test the all-round pressure is 200 kN/m^2 and failure occurs at an added axial stress of 600 kN/m^2 . In the second test all-round pressure is 350 kN/m^2 and failure occurs at an added axial stress of 1050 kN/m^2 . What values of C^* and Φ^* correspond to these results? [05]

SECTION – II

- Q - 4 (a) Derive a relationship between e , G , w & S_r . [04]
 (b) Discuss water transported soils and wind transported soils. [03]
- Q - 5 (a) (1) Explain the field applications of particle size distribution curves. Give the reason of plotting this curve on semi log graph paper. [04]
 (2) Draw particle size distribution curves (only shapes) for well graded soils, poorly graded soils and gap graded soils. [03]
- (b) (1) Enumerate different soil structures and describe honey combed structure and dispersed structure with neat sketches. [04]
 (2) Draw Casagrande's plasticity chart for Unified Soil Classification System. Write formula for A-line. When soil can be classified as ML & MH type? [03]

OR

- Q - 5 (a) (1) The results of the sieve analysis test is given below: [04]

IS Sieve (mm)	10	4.75	2.38	1.18	0.600	0.425	0.212	0.150	0.075	Pan
Mass of soil (gm) retained	28	12	10	7	2	8	6	17	4	5

Total soil mass = 100 g. Plot the particle size distribution curve and find the values of coefficient of curvature and uniformity coefficient. Also identify the type of soil.

- (2) Distinguish between coarse grained soils and fine grained soils. [03]
- (b) (1) Explain different consistency limits and soil states with respect to volume and water content with neat sketch. [04]

(2) Explain textural classification of soil with textural triangle. [03]

Q - 6 (a) Describe the core cutter method to determine field density of soil. [05]

(b) Calculate the value of k of a sample having 6cm height and 50 cm^2 cross-sectional area, if a quantity of water of 430 cm^3 flows down in 10 min under an effective constant head of 40 cm. On oven drying the test specimen weighted 498 g. Assume $G = 2.65$, calculate the discharge velocity and seepage velocity of water. [06]

(c) A horizontal stratified deposit of soil consists of permeability as: Top layer (3 m thick) is $3 \times 10^{-3} \text{ cm/s}$ middle layer (2 m thick) is $6.5 \times 10^{-2} \text{ cm/s}$ and bottom layers (4 m thick) is $7 \times 10^{-4} \text{ cm/s}$. Find the equivalent permeability in vertical direction. [03]

OR

Q - 6 (a) The soil from a pit has natural moisture content 10 % and bulk unit weight 17.65 kN/m^3 . The soil is used for an embankment to be compacted at 18 % moisture content to a dry unit weight of 18.15 kN/m^3 . Determine how many cubic meters of excavation is required for 1 m^3 of compacted embankment? Also calculate the amount of water to be added to 1 m^3 of borrow soil. [05]

(b) Discuss the different factors affecting permeability of soil, in detail. [06]

(c) What is flow net? What are the characteristics of flow net? [03]
